

Logic in AI

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1 Introduction

There are two main goals for this course:

- Introducing a notion of deduction
- Introducing Curry Howard Isomorphism, that connect the notion of computation with the notion of deduction. A proof is just a mathematical object that can be manipulated and transformed.

Definition: Logical System

A logical system has at least three components:

- Syntax (an alphabet, a grammar, a set of well-formed formulas)
- Semantics (in order to link a meaning to the syntax)
- Deductive system (a set of rules to derive new formulas from the existing ones)

Perhaps, in the past you have encountered a logical system like Natural Deduction, tableaux, or sequent calculus.

1.1 Areas of interest

There are several applications of logic in AI, took directly from SOTA (state of art) papers:

- **"Theoritical" Computer Science**: automata, formal languages, computability or recursion theory, complexity theory, etc.
- **Artificial Intelligence**: deduction systems (proof theory), expert systems, NLP, automated proof, multi-agent systems, etc.
- **Programming Languages**: logical programming, resolution functional programming, semantics, type systems, etc.
- **Software Engineering**: program verification, program design, speficications, etc.
- **Hardware**: circuit design verification, etc.

1.2 Roadmap

We will be following a roadmap that is based on the following topics:

1. Propositional logic
   - Natural deduction, with which we will use it to reason about logical principles.
2. First Order Logic
3. Intuitionistic logic
   - $\underbrace{\vdash \varphi \Longleftrightarrow \vDash \varphi}_{\leftarrow}$
   - λ calculus
   - Curry Howard
4. Modal Logics $\boxed{?}, \Diamond$

2 Propositional Logic

We are going to start with propositional logic, which is the simplest logical system.

2.1 Syntax

Propositional logic is composed by:

- Proposition symbols: AT $= \{p_0, p_1, p_2, ...,\} \cup \bot$ ($|\mathbb{N}| = \aleph_0$)
- Connectives:
  - Negation: $\neg\varphi$
  - Conjunction: $\varphi \wedge \psi$
  - Disjunction: $\varphi \vee \psi$
  - Implication: $\varphi \to \psi$
  - Bi-implication: $\varphi \leftrightarrow \psi$
- Auxialiary symbol: $(,)$

The set PROP of propositions is the *smallest* set X with the properties:

1. $p_i \in X (i \in \mathbb{N}), \bot \in X$
2. $\varphi, \psi \in X \Rightarrow (\varphi \wedge \psi), (\varphi \vee \psi), (\varphi \to \psi), (\psi \leftrightarrow \varphi) \in X$
3. $\varphi \in X \Rightarrow (\neg\varphi) \in X$

We need the following induction principles in order to use proof by induction:

Theorem: Induction Principle

Let A be property when $A(\varphi)$ holds for all $\varphi \in$ PROP if:
1. $A(p_i)$ holds for all $i \in \mathbb{N}$ and $A(\bot)$ holds
2. $A(\varphi)$ and $A(\psi)$ hold implies $A(\varphi \wedge \psi)$, $A(\varphi \vee \psi)$, $A(\varphi \to \psi)$, $A(\psi \leftrightarrow \varphi)$ hold
3. $A(\varphi)$ holds implies $A(\neg\varphi)$ holds

Theorem: Definition by Recursion

Let mappings $H_{\boxed{?}} : A^2 \to A$ and $H_\neg : A \to A$ be given and let H_{at} be a mapping from the set of atoms into A, then there exists exactly one mapping $F :$ PROP $\to A$ such that

Theorem: Primitive Recursion

$\exists! f : \mathbb{N} \to \mathbb{N}$ and $h : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ where $c \in \mathbb{N}$ such that $f(0) = c$ and $\forall n \in \mathbb{N}, f(n+1) = H(f(n))$:

- Ω class of satisfying sets Z
  - $(0, c) \in Z, (n, x) \in Z \Rightarrow (s(n), h(n, x))$
  - f intersection of all the sets above
  - f is a set and in Ω

We prove by induction that $\forall n \in \mathbb{N}, \exists! a \in \mathbb{N}$ such that $s \vdash (n, a) \in f$.

- **Base case**: $(0, c) \in f$ by definition. Suppose $\exists d \neq c$ and $f \setminus \{(0, d)\}$ is satisfied and properly contained in f.

Since I am violating the fact that f is the smallest one, this is impossible.

- **Inductive case**: Suppose $\exists! w (i, w) \in f$. By construction of f:

$$(s(i), h(i, w)) \in f$$

suppose $c \neq h(i, w)$ s.t. $(s(i), e) \in f$.

$$f \setminus \{(s(i), e)\} \text{ is satisfied and properly contained in } f$$

$$\Rightarrow \text{impossible}$$

2.2 Semantics

The following two definitions are equal:

Definition: 1

A mapping $v : \text{PROP} \to \{0, 1\}$ is a valuation if:

- $v(p_i) \in \{0, 1\}$ for all $i \in \mathbb{N}$ and $v(\bot) = 0$
- $v(\varphi \wedge \psi) = \min(v(\varphi), v(\psi))$
- $v(\varphi \vee \psi) = \max(v(\varphi), v(\psi))$
- $v(\varphi \to \psi) = \max(1 - v(\varphi), v(\psi))$
- $v(\varphi \leftrightarrow \psi) = 1 \Leftrightarrow v(\varphi) = v(\psi)$
- $v(\neg\varphi) = 1 - v(\varphi)$

Definition: 2

A mapping $v : \text{PROP} \to \{0, 1\}$ is a valuation if:

- $v(p_i) \in \{0, 1\}$ for all $i \in \mathbb{N}$ and $v(\bot) = 0$
- $v(\varphi \wedge \psi) = 1$ iff $v(\varphi) = v(\psi) = 1$
- $v(\varphi \vee \psi) = 1$ iff $v(\varphi) = 1$ or $v(\psi) = 1$
- $v(\varphi \to \psi) = 0$ iff $v(\varphi) = 1$ and $v(\psi) = 0$
- $v(\varphi \leftrightarrow \psi) = 1$ iff $v(\varphi) = v(\psi)$
- $v(\neg\varphi) = 1$ iff $v(\varphi) = 0$

2.3 Tautology

Definition: 3

Let Γ be a set propositions, φ is a tautology if $\forall v$ valuation,

$$\Gamma \vDash \Leftrightarrow \forall v : ([\psi]_v = 1 \quad \text{for all} \quad \psi \in \Gamma) \Leftrightarrow [\varphi]_v = 1$$

2.3.1 Reductio ad absurdum

What does it mean having the bottom as a logical consequence? We have:

$$\Gamma, \neg\alpha \vDash \bot \Rightarrow \Gamma \vDash \alpha$$

Let's provide a valuation for the bottom symbol.

$$\forall v.[|\bot|]_v = 0$$

But if I have something like this:

$$\Delta \vDash \bot$$

Let's try to reason about it using the definition:

$$\forall v.[|\Delta|]_v = 1 \Rightarrow [|\bot|]_v = 1$$

Now let's see the meta-conjuction as a disjunction using this theorem:

$$A \to B := \neg A \vee B$$

So:

$$\underbrace{[|\Delta|]_v \neq 1}_{\forall v.\exists\gamma\in\Delta \text{ s.t. } [|\gamma|]_v=0} \vee \underbrace{[|\bot|]_v = 1}_{\text{false}}$$

This is the meaning of having the bottom has a logical consequence. So:

$$\forall v.\exists\gamma \in \Delta \text{ s.t. } [|\gamma|]_v = 0 \to \Gamma \cup \{\neg\alpha\} \text{ unsat so } \Gamma \vDash \alpha$$

From this, we get the following principles:

- $\neg\alpha \iff \alpha \to \bot$
- $\Gamma, \alpha \vDash \beta \iff \Gamma \vDash \alpha \to \beta$

Let's unwrap this using the rules we have just shown:

$$\Gamma, \neg\alpha \vDash \bot \Longrightarrow \neg\alpha \to \bot$$

So we can derive intuitively the following:

$$\neg\neg\alpha \equiv \alpha$$

We proved the left ($\leftarrow$) direction, but let's now prove the right ($\rightarrow$) direction:

$$\forall v.([|\Gamma|]_v = 1 \Rightarrow [|\alpha|]_v = 1) \iff \forall v.([|\Gamma|]_v = 1 \Rightarrow [|\neg\neg\alpha|]_v = 1)$$

2.4 Natural Deduction

We now consider the persective of the so-called: inference making. The origin of modern notion of inference making? **David Hilbert** proposed a system called **Hilbert-style Deductive System** which is a set of axioms and inference rules. Several mathimaticians understood that this style is not very close to the usual practice of reasoning, so Gentzen proposed **Sequent Calculi** and **Natural Deduction**. In the 60′, Dag Brawitz structured Natural Deduction and wrote a beutiful reference called **Natural Deduction: A Proof Theoritical Study**, which is the very first step towards automated theorem proving.

The structure of Natural Deduction is based on the notion of a *proof tree*, which is a tree where each node is a formula and each edge is an inference rule. The root of the tree is the conclusion and the leaves are the premises.

$$\text{Label}\ \frac{\text{Premise 1} \qquad \text{Premise 2} \qquad \text{Premise 3}}{\text{Conclusion}}\ \text{Rule name}$$

There are usually two kind of rules:

- **Introduction rules**: they introduce a new connective in the conclusion.
- **Elimination rules**: they eliminate a connective from the premises.

For example:

- Elimination rule for implication:

$$\frac{\varphi \to \psi \qquad \varphi}{\psi} \to E$$

- Introduction rule for implication:

$$\frac{\Gamma, \varphi \vdash \psi}{\Gamma \vdash \varphi \to \psi} \to I$$

Let's prove some basic tautologies:

- $A \to A$

$$\frac{[A]^1}{A \to A} \to I^1$$

Let's now add the **Conjunction** rule:

- Introduction rule for conjunction:

$$\frac{\varphi \qquad \psi}{\varphi \wedge \psi} \wedge I$$

- Elimination and introduction rule for conjunction:

$$\frac{\varphi \wedge \psi}{\varphi} \wedge E_1$$

$$\frac{\varphi \wedge \psi}{\psi} \wedge E_2$$

- Introduction and elimination rule for disjunction:

$$\frac{\varphi}{\varphi \vee \psi} \vee I_1$$

$$\frac{\psi}{\varphi \vee \psi} \vee I_2$$

$$\frac{\varphi \vee \psi \qquad \Gamma, \varphi \vdash \chi \qquad \Gamma, \psi \vdash \chi}{\Gamma \vdash \chi} \vee E$$

  Elimination rule for disjunction is a bit more complex, since it requires two premises and a conclusion and it introduce proof by cases.
- Ex falso:

$$\frac{\bot}{\varphi} \bot E$$

  Ex falso is accepted also in intuitionistic logic, but the following is not:
- Reductio ad Absurdum (RAA), a **strong** classical principle

$$\frac{\Gamma, \neg\varphi \vdash \bot}{\Gamma \vdash \varphi}\,\text{RAA}$$

- Tertium non datur:

$$\frac{}{\varphi \vee \neg\varphi}\,\text{TND}$$

  In a classical context you always know if a formula is true or false. In order to prove this, I need to use the RAA rule.

$$\dfrac{\dfrac{\neg\alpha \vee \alpha \qquad \neg(\alpha \vee \neg\alpha)}{\bot}\to E}{\alpha \vee \neg\alpha}$$

Theorem: Pierce's law

Pierce's law is a theorem in propositional logic that states:

$$((P \to Q) \to P) \to P$$

The computational equivalent of Pierce's law is the type of the constructor **call_cc** in Scheme. The derivation of Pierce's law is interesting since we need to use the classical principles:

$$\dfrac{\dfrac{\dfrac{\dfrac{[\alpha]^1 \qquad [\neg\alpha]^2}{\bot}\to E}{\beta}\text{EF}}{\alpha \to \beta}\to I^1 \qquad [(\alpha \to \beta) \to \alpha]^3}{\alpha}\to E \qquad \neg\alpha}{\dfrac{\dfrac{\bot}{\alpha}\text{RAA}^2}{((\alpha \to \beta) \to \alpha) \to \alpha}\to I^3}\to E$$

This is the derivation of the Pierce's law. Redundant rules can be eliminated, in some other deductive systems, but in Natural Deduction we need to keep them so we can close the derivation tree.

2.4.1 Derivation

A derivation φ is a finite tree of formulas where each node is a formula and each edge is an inference rule. The root of the tree is the conclusion and the leaves are the premises.

Definition: Derivability

Let $\mathcal{L}$ be a language and Γ a set of formulas

1. $\Gamma \subseteq \text{WFF}_{\mathcal{L}}$ derives φ ($\Gamma \vdash \varphi$) if there exists $\frac{\Pi}{\varphi}$ s.t. $\mathcal{H}p[\Pi] \subseteq \Gamma$
2. φ is a theorem if $\varnothing \vdash \varphi$

Definition: Eliminability

An inference rule R is eliminable if for each derivation $\frac{\Pi}{\varphi}$ in which R is used there exists another derivation $\frac{\Pi^*}{\varphi}$ in which R is not used and $\mathcal{H}p[\Pi^*] \subseteq \mathcal{H}p[\Pi] \subseteq \Gamma$.

Definition: Equivalence

Let's take two different deductive systems I and J. I and J are equivalent if:

$$\forall \Gamma, \varphi, \Gamma \underset{I}{\vdash} \varphi \iff \Gamma \underset{J}{\vdash} \varphi$$

Exercise for home: Prove that we can sussume / replace in the derivation, EX FALSO by RAA and in the hint is copying the following proof:

Theorem: Example

Let's define $\mathcal{N}$ natural deduction, with this set of rules: {EF, RAA, TND, $\to I/E$, $\wedge I/E$, $\vee I/E$}.

$$\mathcal{N}^* := \mathcal{N} \setminus \{\text{RAA}\} \cup \{\text{TND}\}$$

and $\mathcal{N}$ and $\mathcal{N}^*$ are equivalent.

Proof: $\forall \varphi \ \frac{\Pi}{\varphi}$ in $\mathcal{N}, \exists \frac{\Pi^*}{\varphi}$ in $\mathcal{N}^*$

$$\mathcal{H}p[\Pi] = \mathcal{H}p[\Pi^*]$$

1. Base case: $\Pi \equiv \varphi$ is a derivation of length 1, then $\Pi^* = \Pi$.
2. Inductive step: $\Pi = \frac{\Pi_1 \dots \Pi_k}{\varphi} r$
   1. $r \neq$ RAA by IH

$$\frac{\Pi_1^* \dots \Pi_k^*}{\varphi} r = \Pi^*$$

   2. $r =$ RAA

$$\Pi = \frac{\dfrac{\Pi_1}{\varphi}\,\text{RAA}}{\varphi}$$

$$\Pi^* = \frac{\dfrac{\Pi_1^*}{\varphi}\,\text{TND}}{\varphi}$$

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2.5 Soundness and Completeness

There are two main property about a logic that we would like to connect: syntax and semantics. To do that, we want to have soundness and completeness for the System.

$$\vdash \varphi \iff \models \varphi$$

Where $\implies$ is **soundness** and $\impliedby$ is **completeness**.

There's a nice application of the soundness theorem which is the following one:

Note:

$$\vdash \varphi \Longrightarrow \vDash \varphi \sim \nvDash \varphi \Longrightarrow \nvdash \varphi$$

since

$$A \to B \Leftrightarrow \neg B \to A$$

3 First Order Logic

First Order Logic is a logical system that extends propositional logic by adding quantifiers and variables. It allows us to reason about objects and their properties.

Sometimes I would like to express a formal language. Let's take this language:

$$\mathcal{L} = \{a^n b^n \mid n \geq 0\}$$

We have an alphabet:

$$< r_1, ..., r_n; a_1, ..., a_m; k > \text{ with } r_i \geq 0, a_j > 0$$

- Predicate symbols: $P_1, ..., P_n$ where n is the arity of the predicate.
- Function symbols: $f_1, ..., f_m$ where m is the arity of the function.
- Constant symbols: $c_1, ..., c_k$ where k is the number of constants.
- Variables: $x_1, ..., x_l$ where l is the number of variables. (countably many: $|\text{VAR}| = \aleph_0$)
- Connectives: $\neg, \wedge, \vee, \rightarrow, \leftrightarrow, \bot, \forall, \exists$
- Auxialiary symbols: $(,)$

Definition: Terms

TERM is the smallest set X with the properties:

- $c_i \in X (i \in I)$ and $x_i \in X (i \in N)$.
- $t_1, ..., t_{a_i} \in X \Longrightarrow f_{i(t_1,...,t_{a_i})} \in X$ for $1 \leq i \leq m$.

TERM is our *set of terms.*

Definition: Formulas

FORM is the smalles set X with the properties:

- $\bot \in X; P_i \in X$ if $r_i = 0; t_1, ..., t_{r_i} \in \text{TERM} \Longrightarrow P_{i(t_1,...,t_{r_i})} \in X; t_1, t_2 \in \text{TERM} \Rightarrow t_1 = t_2 \in X$,
- $\varphi, \psi \in X \Longrightarrow (\varphi \wedge \psi), (\varphi \vee \psi), (\varphi \rightarrow \psi), (\psi \leftrightarrow \varphi) \in X$,
- $\varphi \in X \Longrightarrow (\neg\varphi) \in X$,
- $\varphi \in X \Longrightarrow (\forall x_i \varphi), (\exists x_i \varphi) \in X$.

Definition: Induction for terms

Let $A(t)$ be a property of terms. If it holds for t a variable or a constant and if $A(t_1), ...A(t_n) \Rightarrow A(f(t_1, ..., t_n))$ for all function f then $A(t)$ holds for all $t \in \text{TERM}$.

Definition: Induction for formulas

Let $A(\varphi)$ be a property of formulas. If:

- $A(\varphi)$ for atomic φ
- $A(\varphi)$ and $A(\psi)$ implies $A(\varphi \wedge \psi), A(\varphi \vee \psi), A(\varphi \rightarrow \psi), A(\psi \leftrightarrow \varphi)$
- $A(\varphi)$ implies $A(\neg\varphi)$

- $A(\varphi)$ implies $A((\forall x_i)\varphi, A((\exists x_i)\varphi)$ for all i then $A(\varphi)$ holds for all $\varphi \in \text{FORM}$.

Definition: Free variables

The set $\text{FV}(t)$ of free variables of t is defined by:

- $\text{FV}(x_i) := \{x_i\}$, $\text{FV}(\overline{c}_i) := \varnothing$
- $\text{FV}\left(f_{i(t_1,\ldots,t_{a_i})}\right) := \text{FV}(t_1) \cup \ldots \cup \text{FV}(t_n)$

Definition: Free variables for formuals

The set $\text{FV}(t)$ of free variables of t is defined by:

- $\text{FV}(P(t_1, \ldots, t_n)) := \text{FV}(t_1) \cup \ldots \cup \text{FV}(t_n)$
- $\text{FV}(t_1 = t_2) := \text{FV}(t_1) \cup \text{FV}(t_2)$
- $\text{FV}(\varphi \wedge \psi) := \text{FV}(\varphi) \cup \text{FV}(\psi)$, $\text{FV}(\varphi \vee \psi) := \text{FV}(\varphi) \cup \text{FV}(\psi)$, $\text{FV}(\varphi \to \psi) := \text{FV}(\varphi) \cup \text{FV}(\psi)$, $\text{FV}(\psi \leftrightarrow \varphi) := \text{FV}(\varphi) \cup \text{FV}(\psi)$
- $\text{FV}(\forall x_i \varphi) := \text{FV}(\exists x_i \varphi) := \text{FV}(\varphi) - \{x_i\}$

Definition: Closed variables

φ is closed variable if

$$\text{FV}(\varphi) = \varnothing$$

A closed formula is also called a *sentence*. A formula without quantifiers is called *open* formula.

Definition: Subformulas

A subformula can be inductively defined:

- $\text{sub}(\varphi) = \{\varphi\}$
- $\text{sub}(\varphi \odot \psi) = \text{sub}(\varphi) \cup \text{sub}(\psi) \cup \{\varphi \odot \psi\}$ for $\odot \in \{\wedge, \vee, \to, \leftrightarrow\}$
- $\text{sub}(\neg\varphi) = \text{sub}(\varphi) \cup \{\neg\varphi\}$
- $\text{sub}(\forall x_i \varphi) = \text{sub}(\exists x_i \varphi) = \text{sub}(\varphi) \cup \{\forall x_i \varphi, \exists x_i \varphi\}$

Definition: Free and Bound occurrences

An occurence of a variable x in δ is BOUND, if x occurs in $\varphi \in \text{sub}\{\delta\}$ and $\varphi \equiv \{\text{Q}x.\varphi\}$ … FINIRE A CASA

Definition: Substitution

We define substitution of a variable x with a term t in a formula φ as follows:

$$\varphi[t/x]$$

if x is free:

$$\varphi[t/x] = (...t...)$$

if x is bound:

$$\varphi[t/x] = (...x...)$$

So t needs to be free for x in φ.